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Kedist Shegute Dimore

Backend & AI Engineer

Contact

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Location
Addis Ababa, Ethiopia
UAE resident - open to relocate

GitHub
github.com/kedistS

LinkedIn
linkedin.com/in/kedist-shegute

Skills

Languages
Python · Java · C++ · JavaScript · Dart · PHP

Backend & APIs
Python · FastAPI · REST APIs · Node.js

Data & DevOps
PostgreSQL · MySQL · MongoDB · Docker · CI/CD

AI & Graph
Neo4j · BioCypher · SPMiner · TensorFlow · Keras

Profile

Backend & AI Engineer with 3+ years of professional experience at iCog Labs, building Python/FastAPI APIs, data pipelines, Dockerized services, Neo4j knowledge graph systems, and LLM/graph-mining tools. B.Sc. Computer Science graduate from Addis Ababa University, 2024. UAE residency visa holder with a twofour54 Software Developer license, available for UAE, remote, or hybrid backend and AI engineering roles.

Experience

Team Leader, NeuroGraph AI Assistant

iCog Labs | October 2025 to April 2026

- Led a four-service AI platform from proposal to deployment, combining Python/FastAPI backend services, neural subgraph mining, LLM interpretation, and graph visualization.
- Designed service responsibilities, API workflows, Docker deployment, and team delivery plans for querying and explaining graph motifs across biological networks.
- Managed code reviews, implementation coordination, and production handoff while keeping backend, AI, and interface components aligned.

Computer Scientist

iCog Labs | April 2023 to Present

- Built and maintained backend/data pipelines for a large biomedical knowledge graph integrating 34 human sources and 11 Drosophila sources across genes, proteins, regulatory elements, and diseases.
- Improved graph import and refresh workflows, reducing manual maintenance and enabling faster, more reliable updates for large Neo4j/BioCypher datasets.
- Built automated source-release detection, schema documentation, and affected-area CI/CD checks to make data updates easier to test and review.
- Worked across Python, APIs, databases, Docker, graph tooling, and production-oriented research systems.

Graph Pattern Mining & Biological Discovery

iCog Labs / Rejuve.Bio

- Applied neural subgraph mining to a BioCypher human biological KG with 150,619 nodes and approximately 20M edges, sampling 20,000 ego-network neighborhoods.
- Characterized 18 patterns across 11,311 discovered KG instances and 10,138 unique entities, separating dense PPI subgraphs, flexible annotation fans, and cross-domain motifs.
- Analyzed TFLink and STRING results, including 503 recurrent regulatory pattern instances and 14,877 STRING PPI pattern instances across 18 motif types.

Impact

20M edge BioCypher KG analyzed	**11,311** KG pattern instances discovered
14,877 STRING PPI pattern instances analyzed	**503** TFLink regulatory motifs reported

License

UAE work-ready
Residency visa holder
twofour54 Software Developer
License #3767

Languages

English	Advanced
Amharic	Native
Korean	Reading/listening

Interests

- Artificial Intelligence
- Graph Science
- Web Development
- Technical Writing

Tools

Engineering
Git · Linux · Selenium · WordPress · CI/CD

Graph & Bio
Neo4j GDS · SPMiner · BioCypher · NetworkX · OWL/RDF

Backend Engineering

Backend APIs, Data Pipelines & Automation

- Built Python/FastAPI service workflows, Dockerized AI assistant services, affected-area CI/CD testing, and generated schema documentation from source code.
- Worked with PostgreSQL, MySQL, MongoDB, Neo4j, Linux, Git, and API-connected application workflows across professional and project work.

Personal Projects

Coffee Disease Detection and Classification System

Final Year Project | 2023 to 2024

Built an end-to-end ML and backend workflow for coffee plant disease detection, from data collection and labeling through model training, evaluation, API integration, and frontend delivery.

Campus PC Authentication System

Personal Project | 2023

Built a Flutter mobile app for university device registration, generating QR code tags that security staff scan to verify device ownership.

Lounge Management System

Personal Project | 2022 to 2023

Developed a Java Swing desktop app for campus lounge inventory, customer ordering, and real-time order tracking.

Research

Neural Subgraph Mining Analysis of a Human Biological Knowledge Graph

Analysis Report | BioCypher KG, 2026

Analyzed a 150,619-node, ~20M-edge KG integrating GO, STRING, TFLink, GENCODE, UniProt, and GAF data; characterized 11,311 discovered instances across 18 motifs.

Hierarchical Regulatory Patterns in Human Transcription Factor Networks

Manuscript | iCog Labs / Rejuve.Bio, 2026

Analyzed the TFLink network and reported 503 recurrent hierarchical pattern instances across 374 unique genes, including AGPAT5 and IMPACT as frequent terminal convergence points.

Structural Motif Analysis in Protein-Protein Interaction Networks

Manuscript | iCog Labs / Rejuve.Bio, 2026

Analyzed STRING PPI mining results with 14,877 pattern instances across 18 motif types, highlighting modules in immune signaling, chromatin regulation, mitochondria, DNA replication, and transport.

Education

B.Sc. in Computer Science

Graduated: July 2024

Certifications

Coursera Specializations

- Supervised Machine Learning: Regression and Classification
- Neural Networks and Deep Learning
- Mathematics for Machine Learning: Linear Algebra